Percentage Change

Exam code: 0607

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Written by: Jamie Wood
Reviewed by: Dan Finlay
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Percentage change

How do I find a percentage change?

- The **multiplier** that was used for a **percentage change** can be found using the formula:
  - $m = \frac{\text{Amount after}}{\text{Amount before}}$
- The value of m corresponds to the **multiplier** for the percentage change
  - **1.05** corresponds to an **increase by 5%**
  - **0.75** corresponds to a **decrease by 25%**
- **Alternatively** you can use the formula:
  - Percentage Change = $\frac{\text{After} - \text{Before}}{\text{Before}} \times 100$
  - A **positive** value is a percentage **increase**
    - An answer of 12 means an increase of 12%
  - A **negative** value is a percentage **decrease**
    - An answer of -28 means a decrease of 28%

How do I find a percentage profit or loss?

- Similar strategies to the above can be used to find the percentage profit or loss
- Shops buy or produce items at a "cost price" and sell them at a "selling price"
- Using a **multiplier** method:
  - $m = \frac{\text{Selling Price}}{\text{Cost Price}}$
  - **1.05** corresponds to a **5% profit**
  - **0.75** corresponds to a **25% loss**
- **Alternatively** you can use the formula:
  - Percentage Profit = $\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}} \times 100$
  - A **positive** value is a **profit**
    - An answer of 12 means a 12% profit
  - A **negative** value is a **loss**
    - An answer of -28 means a 28% loss

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Use "common sense" to check your answer!

If an item is sold for more than it was bought for, you are expecting a profit, not a loss.

Worked Example

The number of students in a school changes from 250 to 310. Describe the percentage change in number of students.

Answer:

Method 1

Use the formula $m = \frac{\text{Amount after}}{\text{Amount before}}$

$$\frac{310}{250} = 1.24$$

This multiplier corresponds to an increase of 24%

A percentage increase of 24%

Method 2

Use the formula Percentage Change = $\frac{\text{After} - \text{Before}}{\text{Before}} \times 100$

$$\frac{310 - 250}{250} \times 100 = 24$$

The value is positive, so this is a percentage increase

A percentage increase of 24%

Worked Example

Sophie purchases a car for £8000 and sells it several years later for £5600. Describe the percentage profit or loss on the car.

Answer:

Method 1

Use the formula $m = \frac{\text{Selling Price}}{\text{Cost Price}}$

$$\frac{5600}{8000} = 0.7$$

This means the selling price was 70% of the cost price, so a loss of 30%

A loss of 30%

Method 2

Use the formula Percentage Profit =

$$\frac{5600 - 8000}{8000} \times 100 = -30$$

The value is negative, so this is a percentage loss

A loss of 30%

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